

UKOH FAITH PRINCE

Geography (Remote Sensing) | GIS & Remote Sensing | Environmental Data Analysis

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RESEARCH PROFILE

Early-career Geography (Remote Sensing) researcher with practical experience in GIS, Earth Observation, environmental research, spatial analysis, field data collection and quantitative data interpretation. Hands-on experience with ArcGIS Pro, Landsat, Sentinel-2 and NASA POWER climate datasets, with research interests spanning land-use/land-cover dynamics, urban heat, rainfall variability, flood susceptibility and environmental risk assessment. Interested in applying geospatial and Earth Observation methods to locally relevant environmental challenges and evidence-based decision-making.

EDUCATION

Federal University of Technology, Minna (FUTMINNA), Minna, Niger State, Nigeria

B.Tech. Geography (Remote Sensing) — Degree requirements completed; awaiting award, 2026

Relevant coursework: Geographic Information Systems, Remote Sensing, Climatology, Hydrology, Environmental Management, Natural Resources Evaluation, Spatial Analysis, Digital Image Processing and Research Methods.

RESEARCH & PROJECT EXPERIENCE

Influence of Land Use and Land Cover Dynamics on Urban Heat Island Formation in Minna and Its Environs, Nigeria

Undergraduate project examining LULC dynamics, vegetation conditions, Land Surface Temperature (LST) and Urban Heat Island (UHI) formation using Remote Sensing and GIS. Work includes Landsat-based analysis, NDVI, LST, spatial interpretation and change analysis for 2005, 2015 and 2025.

Flood Susceptibility Assessment of Makurdi Metropolis, Benue State, Nigeria

GIS and Remote Sensing-based assessment integrating Sentinel-1/Sentinel-2, DEM, slope, drainage and LULC variables to identify spatial patterns of flood susceptibility. The work demonstrates application of Earth Observation and spatial analysis to flood-risk management, urban planning and disaster preparedness.

Assessment of Rainfall Variability and Urban Flooding in Karu LGA, Nasarawa State, Nigeria

Analyzed long-term rainfall variability and flood patterns using climate and spatial datasets; applied GIS and quantitative methods to assess flood-risk patterns and environmental conditions; produced thematic maps, spatial outputs and statistical summaries.

Socioeconomic Impact of Tagwai Dam on Surrounding Settlements, Niger State, Nigeria

Designed survey instruments and supported field data collection; applied spatial analysis to examine relationships between the dam, surrounding settlements and socioeconomic conditions; contributed maps, tables and written research analysis.

TECHNICAL SKILLS

GIS & Spatial Analysis	ArcGIS Pro, Google Earth; thematic mapping, spatial data management, spatial analysis
Remote Sensing & Earth Observation	Landsat, Sentinel-1, Sentinel-2, image classification, LULC analysis, NDVI, LST
Environmental & Climate Data	NASA POWER, rainfall variability analysis, environmental data interpretation, DEM, slope and drainage analysis
Research & Data	Microsoft Excel, quantitative analysis, questionnaire design, field data collection, trend analysis, data visualization
Reporting & Communication	Research reports, thematic maps, charts, spatial visualization, presentations and infographics

RESEARCH INTERESTS

Earth Observation for environmental monitoring; urban climate and heat exposure; LULC change; flood susceptibility and urban hydrology; rainfall and climate variability; spatial modelling; environmental risk assessment; GIS-based decision support; and climate-resilient urban planning.

SELECTED RESEARCH METHODS & APPLICATIONS

- Satellite image processing and interpretation using Landsat and Sentinel datasets.
- LULC classification and change-detection workflows.
- NDVI-based vegetation assessment and LST/UHI analysis.
- GIS-based terrain and drainage analysis using DEM, slope and drainage variables.
- Integration of raster and vector datasets for spatial susceptibility assessment.
- Google Earth Engine, ArcGIS Pro and Google Earth workflows for Earth Observation and spatial outputs.
- Quantitative interpretation of environmental and climate datasets and presentation through maps, charts and reports.

LEADERSHIP, FACILITATION & COMMUNITY ENGAGEMENT

Speech Africa Hub — Lead

Support and coordinate learning and development activities focused on communication, public speaking and personal development; facilitate virtual learning sessions and coordinate participants and facilitators.

Nigeria Fellowship of Evangelical Students (NIFES), Minna Zone — Zonal Prayer Secretary | Leadership & Training Volunteer

Approximately three years of involvement in student leadership, training and development initiatives; coordinate schedules, communication and activities involving students, leaders and team members.

Facilitation & Communication

Facilitate student-focused sessions on public speaking, project presentation, communication and academic development; engage participants through presentations, discussions and interactive learning activities.

PROFESSIONAL DEVELOPMENT

GIS and Remote Sensing Practical Training • Data Visualization and Spatial Analysis Workshops • Ongoing development in Leadership, Communication, Public Speaking and Facilitation • Ongoing learning in Environmental Data Analysis and Digital Mapping

IEEE GRSS ALIGNMENT

The IEEE GRSS West Africa Summer School aligns directly with my academic and research direction in Remote Sensing, GIS and Earth Observation. I seek to strengthen my technical capacity in geospatial science and apply advanced methods to environmental challenges in West African cities, particularly urban heat, flooding and climate-related risk.

CORE COMPETENCIES

Remote Sensing • GIS • Earth Observation • Spatial Analysis • LULC Classification • NDVI • LST • UHI • Flood Susceptibility • Environmental Data Analysis • Field Data Collection • Data Visualization • Research Reporting

REFERENCES

Available upon request.